

UNIVERSITY COLLEGE OF ENGINEERING NAGERCOIL

(ANNA UNIVERSITY CONSTITUENT COLLEGE)

KONAM, NAGERCOIL – 629 004



RECORD NOTE BOOK

CCS334-BIG DATA ANALYTICS

REGISTER NO:

UNIVERSITY COLLEGE OF ENGINEERING NAGERCOIL

(ANNA UNIVERSITY CONSTITUENT COLLEGE)

KONAM, NAGERCOIL – 629 004



Register No:

*Certified that, this is the Bonafide record of work done by
Mr./Ms. of VI Semester
in Computer Science and Engineering of this college, in the
CCS334-BIG DATA ANALYTICS during academic year 2025-2026 in
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Staff-in-charge

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EX NO:1

**DOWNLOADING AND INSTALLING HADOOP
UNDERSTANDING DIFFERENT HADOOP MODES,
STARTUP SCRIPTS, CONFIGURATION FILES**

DATE:

AIM:

To Downloading and installing Hadoop; Understanding different Hadoop modes.

Startup scripts, Configuration files.

PREREQUISITES TO INSTALL HADOOP ON WINDOWS:

VIRTUAL BOX (For Linux): it is used for installing the operating system on it.

OPERATING SYSTEM: You can install Hadoop on Windows or Linux based operating systems. Ubuntu and CentOS are very commonly used.

Java: U need to install java 8 on your system

Hadoop: You required Hadoop latest version

1. Install Java

Java JDK Link to download

here Extract and install Java

in C:\Java Open cmd and

type -> javac -version

2. Download Hadoop

Hadoop Link to download:

<https://www.apache.org/dyn/closer.cgi/hadoop/common/hadoop-3.3.0/hadoop-3.3.0.tar.gz>

Extract to C:\Hadoop

3. Set the path JAVA_HOME Environment variable

4. Set the path HADOOP_HOME Environment variable

5. Configure the required files in the

C:\hadoop\etc as follows: # For core-site.xml

<configuration>

<property>

<name>fs.defaultFS</name>

<value>hdfs://localhost:9000</value>

</property>

</configuration>

For hdfs-site.xml or https-site.xml

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property><property>
<name>dfs.namenode.name.dir</name>
<value>\hadoop-3.3.0\data\namenode</value>
</property><property>
<name>dfs.datanode.data.dir</name>
<value>\hadoop-3.3.0\data\datanode</value>
</property>
</configuration>
```

For mapred-site.xml

```
<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
</configuration>
```

For yarn-site.xml

```
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property><property>
<name>yarn.timeline-services.enabled</name>
<value>>false</value>
```

</property>

</configuration>

6. Create the following folders:

Create folder “data” under “C:\Hadoop”

Create folder “datanode” under “C:\Hadoop-

3.3.0\data” Create folder “namenode” under

“C:\Hadoop-3.3.0\data”

7. Edit file hadoop-env.cmd:

C:/Hadoop-3.3.0/etc/hadoop/hadoop-env.cmd by closing the command line “JAVA HOME=%JAVA HOME%” instead of set ;

```
@rem The java implementation to use. Required.
set JAVA_HOME=C:\Java\jdk-1.8
```

Also add these line ad the end of the file:

```
set HADOOP_CLASSPATH=%HADOOP_CLASSPATH%;%HADOOP_HOME%\share
\hadoop\yarn\timelineservice\*
```

8. Hadoop Configurations:

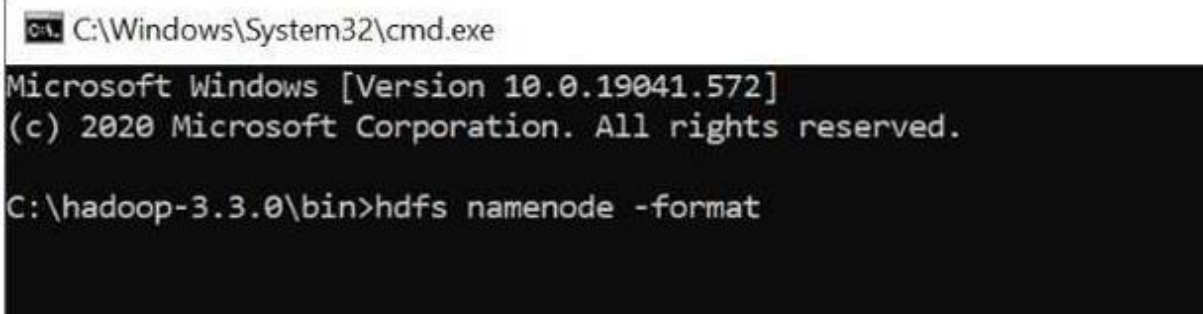
Download [https://github.com/s911415/apache-hadoop-](https://github.com/s911415/apache-hadoop-3.1.0-winutil)

3.1.0-winutil Copy folder bin and replace existing bin folder

in C:\hadoop-3.3.0\bin.

9. Format the NameNode:

Open cmd and type command “hdfs namenode –format”



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19041.572]
(c) 2020 Microsoft Corporation. All rights reserved.

C:\hadoop-3.3.0\bin>hdfs namenode -format
```

10. Start Hadoop:

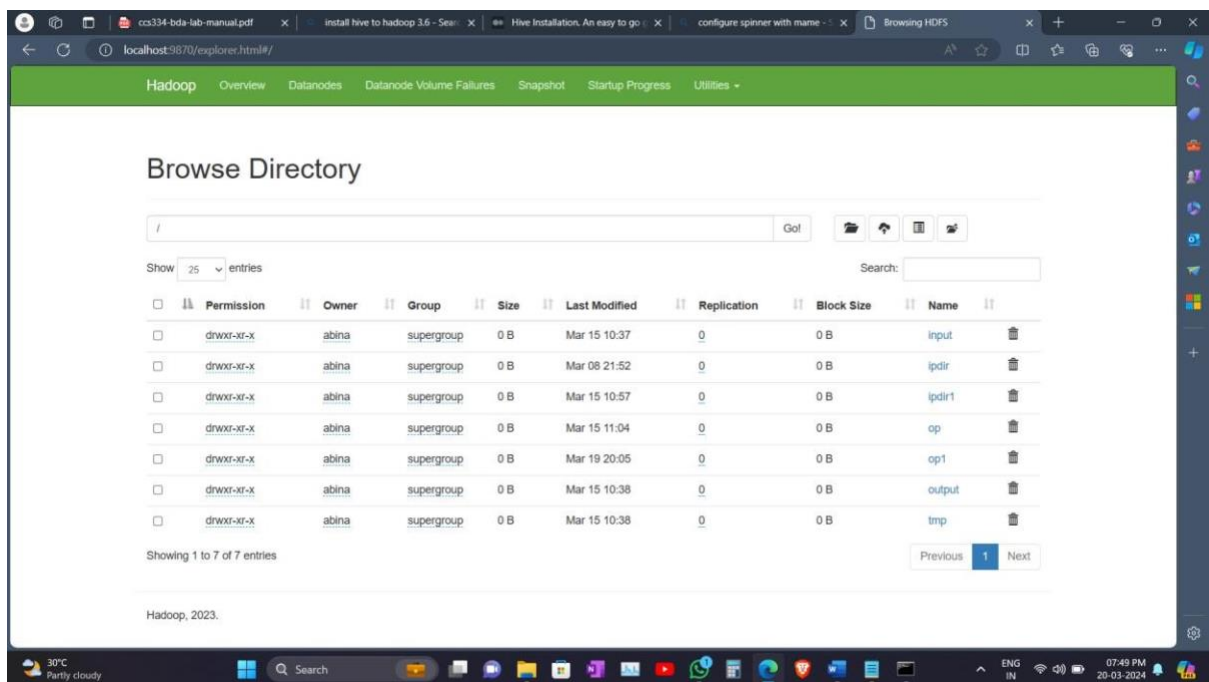
Open cmd and type command “start-all.cmd”

Then verify the following windows appear without any warnings and errors.

```
Apache Hadoop Distribution - hadoop namenode
20/11/08 00:21:09 INFO Apache Hadoop Distribution - hadoop datanode
20/11/08 00:21:09 INFO Apache Hadoop Distribution - yarn ResourceManager
20/11/08 00:21:09 INFO Apache Hadoop Distribution - yarn NodeManager
20/11/08 00:21:09 INFO the scope "Singleton"
20/11/08 00:21:09 INFO com.sun.jersey.guice.spi.container.GuiceComponentProviderFactory getComponentProvider
20/11/08 00:21:09 INFO: Binding org.apache.hadoop.yarn.webapp.GenericExceptionHandler to GuiceManagedComponentProvider with the scope "Singleton"
20/11/08 00:21:10 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:10 INFO com.sun.jersey.guice.spi.container.GuiceComponentProviderFactory getComponentProvider
20/11/08 00:21:10 INFO: Binding org.apache.hadoop.yarn.server.nodemanager.webapp.NMWebServices to GuiceManagedComponentProvider with the scope "Singleton"
20/11/08 00:21:10 INFO mortbay.log: Started HttpServer2$SelectChannelConnectorWithSafeStartup@0.0.0.0:8042
20/11/08 00:21:10 INFO webapp.WebApps: Web app node started at 8042
20/11/08 00:21:10 INFO nodemanager.NodeStatusUpdaterImpl: Node ID assigned is : DESKTOP-475FCII:61797
20/11/08 00:21:10 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8031
20/11/08 00:21:10 INFO nodemanager.NodeStatusUpdaterImpl: Sending out 0 NM container statuses: []
20/11/08 00:21:10 INFO nodemanager.NodeStatusUpdaterImpl: Registering with RM using containers: []
20/11/08 00:21:10 INFO security.NMContainerTokenSecretManager: Rolling master-key for container-tokens, got key with id 528277285
20/11/08 00:21:10 INFO security.NMTokenSecretManagerInNM: Rolling master-key for container-tokens, got key with id 28336
20/11/08 00:21:10 INFO nodemanager.NodeStatusUpdaterImpl: Registered with ResourceManager as DESKTOP-475FCII:61797 with total resource of <memory:8192, vCores:8>
20/11/08 00:21:10 INFO nodemanager.NodeStatusUpdaterImpl: Notifying ContainerManager to unblock new container-requests
20/11/08 00:21:13 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:16 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:19 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:22 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:25 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:28 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:31 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
20/11/08 00:21:34 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
```

11. Open localhosts:

Localhost:9870



Localhost:8088

Localhost:8088

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources
0	0	0	0	0	<memory:0 B, vCores:0>

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes
1	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>

Showing 20 entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers
No data available in table												

Showing 0 to 0 of 0 entries

Result:

To Downloading and installing Hadoop; Understanding different Hadoop modes.

Startup scripts, Configuration files

EX NO:2

DATE:

HADOOP IMPLEMENTATION OF FILE MANAGEMENT TASKS SUCH AS ADDING FILES AND DIRECTORIES, RETRIEVING FILES AND DELETING FILES

AIM:

To implement the following file management tasks in Hadoop:

1. Adding files and directories
2. Retrieving files
3. Deleting Files

PROCEDURE:

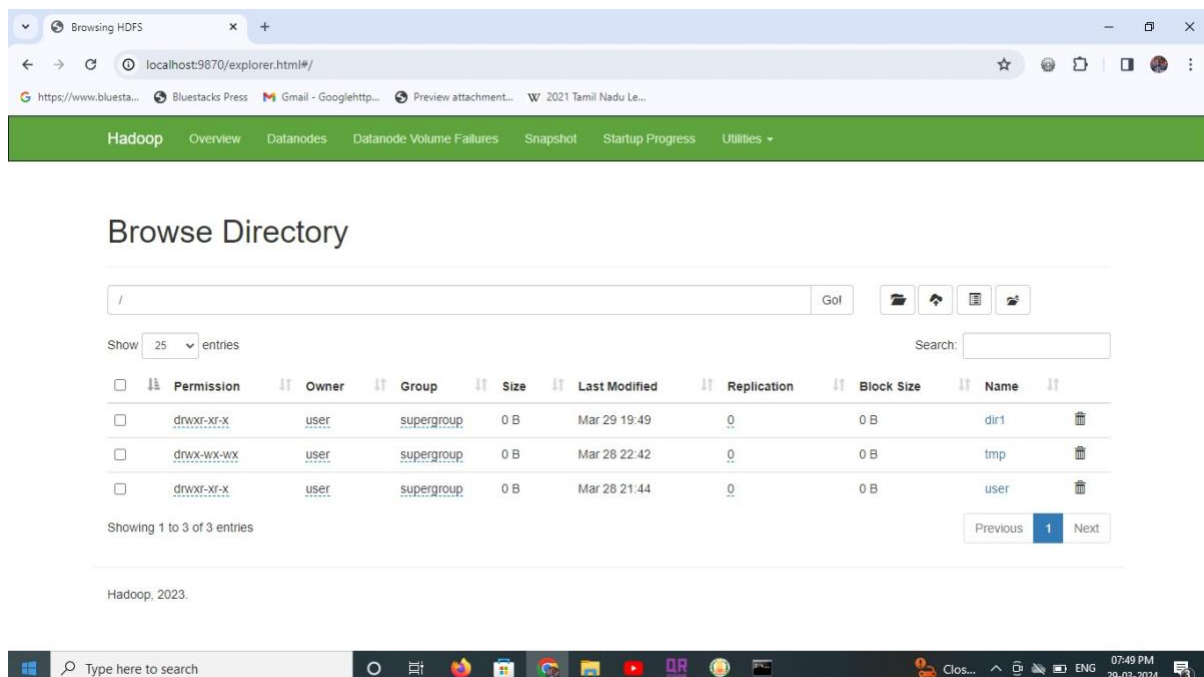
1.Create a directory in HDFS at given path:

Usage:

`hadoop fs -mkdir <paths>` Example:

`hadoop fs -mkdir /user/saurzcode/dir1 /user/saurzcode/dir2`

```
C:\hadoop\sbin>hadoop fs -mkdir /dir1  
C:\hadoop\sbin>
```



The screenshot shows a web browser window titled "Browsing HDFS" with the address bar displaying "localhost:9870/explorer.html#/". The browser tabs include "https://www.bluesta...", "Bluestacks Press", "Gmail - Googlehttp...", "Preview attachmen...", and "2021 Tamil Nadu Le...". The Hadoop web interface has a green navigation bar with links: "Hadoop", "Overview", "Datanodes", "Datanode Volume Failures", "Snapshot", "Startup Progress", and "Utilities". The main content area is titled "Browse Directory" and shows a directory listing for the root path "/". The listing includes a search bar, a "Show 25 entries" dropdown, and a table with columns: "Permission", "Owner", "Group", "Size", "Last Modified", "Replication", "Block Size", and "Name". The table lists three entries: "dir1" (permissions drwxr-xr-x, owner user, group supergroup, size 0 B, last modified Mar 29 19:49, replication 0, block size 0 B), "tmp" (permissions drwx-wx-wx, owner user, group supergroup, size 0 B, last modified Mar 28 22:42, replication 0, block size 0 B), and "user" (permissions drwxr-xr-x, owner user, group supergroup, size 0 B, last modified Mar 28 21:44, replication 0, block size 0 B). At the bottom of the table, it says "Showing 1 to 3 of 3 entries" and "Hadoop, 2023."

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwxr-xr-x	user	supergroup	0 B	Mar 29 19:49	0	0 B	dir1
drwx-wx-wx	user	supergroup	0 B	Mar 28 22:42	0	0 B	tmp
drwxr-xr-x	user	supergroup	0 B	Mar 28 21:44	0	0 B	user

2. List the contents of a directory:

Usage:

`hadoop fs -ls <args>`

Example:

`hadoop fs -ls /user/saurzcode`

```
C:\Windows\System32\cmd.exe
C:\hadoop\sbin>hadoop fs -touchz /dir1/1.txt

C:\hadoop\sbin>hadoop fs -ls /dir1
Found 1 items
-rw-r--r--    1 user supergroup          0 2024-03-29 19:51 /dir1/1.txt

C:\hadoop\sbin>
```

3. Upload and download a file in HDFS:

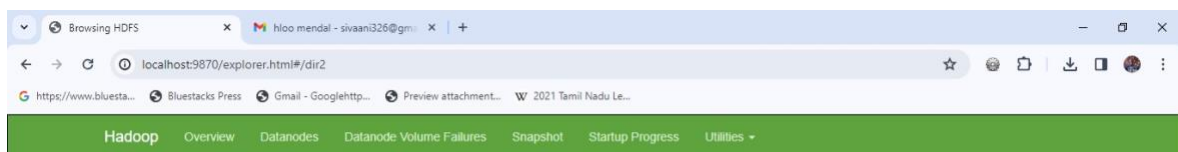
Upload: `hadoop fs -put`:

Copy single src file, or multiple src files from local file system to the Hadoop data file system Usage: `hadoop fs -put <localsrc>...<HDFS_dest_Path>`Example:

`hadoop fs -put /home/saurzcode/Samplefile.txt /user/saurzcode/dir3/`

```
C:\hadoop\sbin>hadoop fs -put C:\Users\user\b.py /dir2

C:\hadoop\sbin>
```



Browse Directory

/dir2

Go!



Show

25

entries

Search:

☐	 Permission	 Owner	 Group	 Size	 Last Modified	 Replication	 Block Size	 Name	
☐	-rw-r--r--	user	supergroup	0 B	Mar 29 19:58	1	128 MB	1.txt	
☐	-rw-r--r--	user	supergroup	2.61 KB	Mar 29 22:13	1	128 MB	b.py	

Showing 1 to 2 of 2 entries

Previous

1

Next

Hadoop, 2023.



Adding files to a directory:

hadoop fs -touchz:

Adds files to a

directory. Usage:

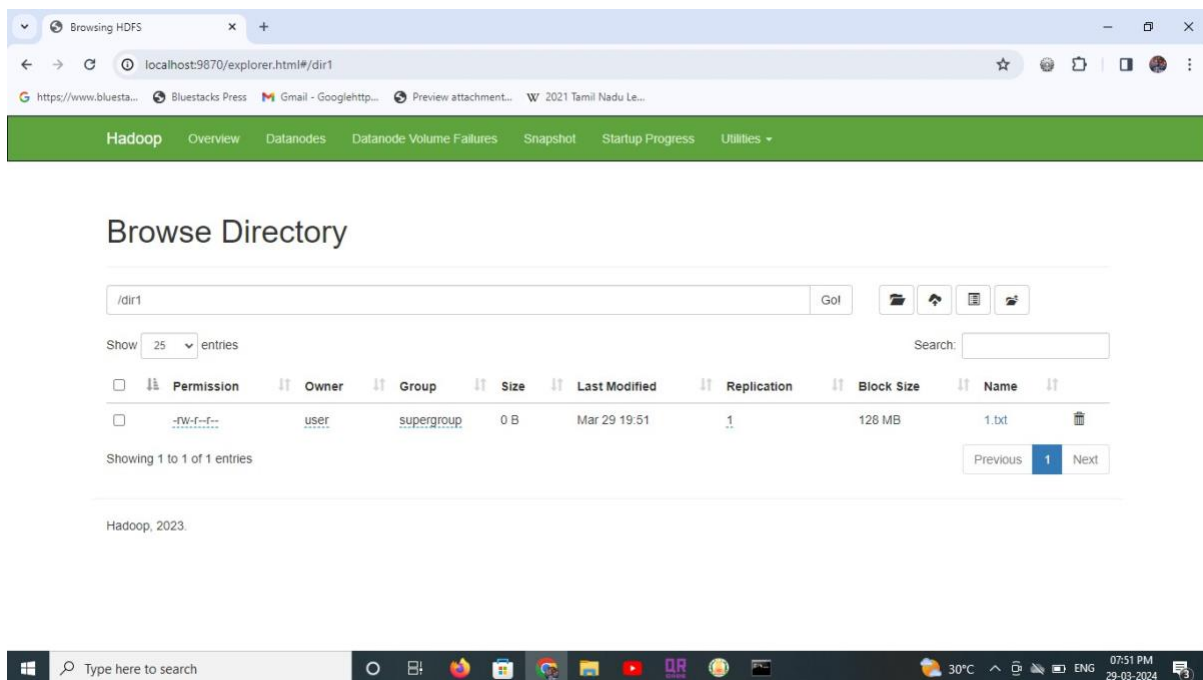
hadoop fs -touchz

/directory_name/file

Example:

hadoop fs -touchz /dir1/1.txt

```
C:\Windows\System32\cmd.exe
C:\hadoop\sbin>hadoop fs -touchz /dir1/1.txt
C:\hadoop\sbin>_
```



4. Remove a file or directory in HDFS.

Remove files specified as argument. Deletes directory only when it is empty
Usage: `hadoop fs -rm <arg>` Example:

`hadoop fs -`

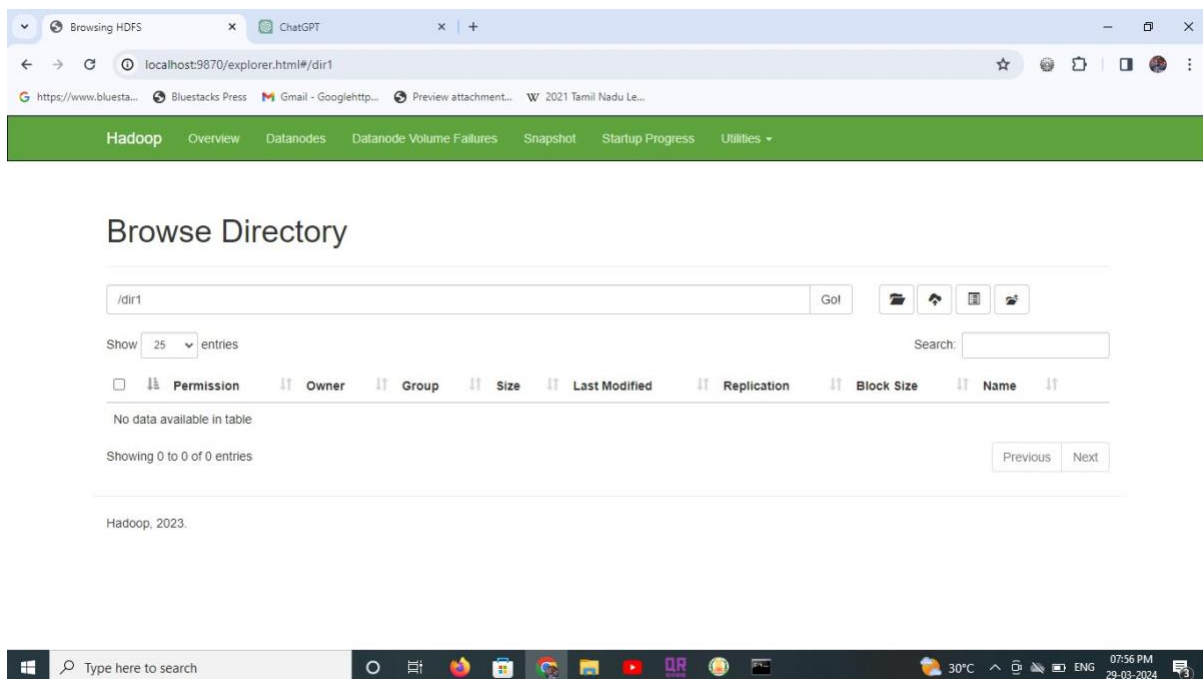
`rm/user/saurzcode/dir1/abc.txt`

Recursive version of delete.

Usage:

`hadoop fs -rm <arg>` Example: `hadoop fs -rmr/user/saurzcode/`

```
C:\Windows\System32\cmd.exe
C:\hadoop\sbin>hadoop fs -rm /dir1/1.txt
Deleted /dir1/1.txt
C:\hadoop\sbin>
```



5. Move file from source to destination.

Note :- Moving files across filesystem is not permitted.

Usage :

`hadoop fs -mv <src> <dest>`

Example:

```
hadoop fs -mv /user/saurzcode/dir1/abc.txt /user/saurzcode/ dir2
```

6.Display last few lines of a file.

Similar to tail command in Unix.

Usage :

hadoop fs -tail <path[filename]> Example:

```
hadoop fs -tail /user/saurzcode/dir1/abc.txt
```

7. Display the aggregate length of a file.

Usage :

hadoop fs -du <path> Example:

```
hadoop fs -du /user/saurzcode/dir1/abc.txt
```

RESULT:

Thus, the Hadoop Implementation of file management tasks, such as Adding files and directories, retrieving files and Deleting files is executed successfully.

AIM:

To implement matrix multiplication using mapreduce in Hadoop.

PROGRAM:**MatrixMultiply.java**

```
import java.io.*;
import org.apache.hadoop.conf.*;
import org.apache.hadoop.fs.*;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.*;
import org.apache.hadoop.mapreduce.lib.output.*;

public class MatrixMultiply {

    public static class M extends Mapper<LongWritable, Text, Text, Text> {
        public void map(LongWritable k, Text v, Context c) throws IOException,
            InterruptedException {
            String[] e = v.toString().split(",");
            String type = e[0], row = e[1], col = e[2], val = e[3];
            if(type.equals("A")) c.write(new Text(row+","+col), new Text("A,"+val));
            else c.write(new Text(row+","+col), new Text("B,"+val));
        }
    }

    public static class R extends Reducer<Text, Text, Text, IntWritable> {
        public void reduce(Text k, Iterable<Text> vals, Context c) throws IOException,
            InterruptedException {
            int a=0, b=0;
            for(Text t: vals){
                String[] s=t.toString().split(",");
                if(s[0].equals("A")) a=Integer.parseInt(s[1]);
                else b=Integer.parseInt(s[1]);
            }
            c.write(k, new IntWritable(a*b));
        }
    }

    public static void main(String[] args) throws Exception {
        Job job = Job.getInstance(new Configuration(), "matrixmul");
        job.setJar("m1.jar");
        job.setJarByClass(MatrixMultiply.class);

        job.setMapperClass(M.class);
        job.setReducerClass(R.class);
    }
}
```

```

        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(Text.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job,new Path(args[0]));
        FileOutputFormat.setOutputPath(job,new Path(args[1]));

        System.exit(job.waitForCompletion(true)?0:1);
    }
}

```

PROCEDURE:

STEP 1: START HADOOP

Open CMD as Administrator

```
cd C:\hadoop-3.3.0\sbin
```

```
start-dfs.cmd
```

```
start-yarn.cmd
```

STEP 2: CREATE PROJECT FOLDER

```
mkdir C:\HadoopWork
```

```
mkdir C:\HadoopWork\src
```

```
cd C:\HadoopWork\src
```

STEP 3: WRITE INPUT FILE

create a input file called sam.txt:

```
A,0,0,2
```

```
B,0,0,3
```

```
A,1,1,4
```

```
B,1,1,5
```

STEP 4: WRITE JAVA PROGRAM

Create MatrixMultiply.java in the same folder.

STEP 5: COMPILE

```
javac -classpath "C:\hadoop-3.3.0\share\hadoop\common\*;C:\hadoop
3.3.0\share\hadoop\common\lib\*;C:\hadoop-3.3.0\share\hadoop\mapreduce\*;C:\hadoop
3.3.0\share\hadoop\mapreduce\lib\*" MatrixMultiply.java
```

STEP 6: CREATE JAR FILE

```
jar cf m1.jar MatrixMultiply*.class
```

STEP 7: CREATE HDFS INPUT DIRECTORY

```
hdfs dfs -mkdir /in
```

```
hdfs dfs -put sam.txt /in
```

```
hdfs dfs -ls /in
```

STEP 8: RUN MAPREDUCE JOB

```
hadoop jar m1.jar MatrixMultiply /in /out
```

```
C:\HadoopWork\src>hadoop jar m1.jar MatrixMultiply /in /out
2026-03-20 12:46:45,336 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-20 12:46:46,012 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-20 12:46:46,034 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/SriJainthu/.staging/job_1773989457437_0003
2026-03-20 12:46:46,337 INFO input.FileInputFormat: Total input files to process : 1
2026-03-20 12:46:46,424 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-20 12:46:46,557 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773989457437_0003
2026-03-20 12:46:46,557 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-20 12:46:46,743 INFO conf.Configuration: resource-types.xml not found
2026-03-20 12:46:46,743 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-20 12:46:46,808 INFO impl.YarnClientImpl: Submitted application application_1773989457437_0003
2026-03-20 12:46:46,879 INFO mapreduce.Job: The url to track the job: http://Jainthu-Pc:8088/proxy/application_1773989457437_0003/
2026-03-20 12:46:46,880 INFO mapreduce.Job: Running job: job_1773989457437_0003
2026-03-20 12:46:57,088 INFO mapreduce.Job: Job job_1773989457437_0003 running in uber mode : false
2026-03-20 12:46:57,088 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-20 12:47:03,223 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-20 12:47:11,301 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-20 12:47:12,329 INFO mapreduce.Job: Job job_1773989457437_0003 completed successfully
2026-03-20 12:47:12,563 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=46
    FILE: Number of bytes written=554621
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
```

STEP 9: VIEW OUTPUT

```
hdfs dfs -ls /out
```

```
hdfs dfs -cat /out/part-r-00000
```

```
C:\HadoopWork\src>hdfs dfs -cat /out/part-r-00000
0,0      6
1,1      20
```

RESULT:

Thus the implementation of matrix multiplication using map reduce was executed successfully.

EX NO:4

DATE:

RUN A BASIC WORD COUNT MAP REDUCE PROGRAM TO UNDERSTAND MAP REDUCE PARADIGM

AIM:

To run a basic Word Count Map Reduce program to understand Map Reduce Paradigm.

PREREQUISITES:

VIRTUAL BOX (For Linux): it is used for installing the operating system on it.

OPERATING SYSTEM: You can install Hadoop on Windows or Linux based operating systems. Ubuntu and CentOS are very commonly used.

JAVA: You need to install the Java 8 package on your system.

HADOOP: You require Hadoop latest version

PROGRAM:

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class WordCount {
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();
        public void map(LongWritable key, Text value, Context context) throws IOException,
        InterruptedException {
            String line = value.toString();
            StringTokenizer tokenizer = new StringTokenizer(line);
            while (tokenizer.hasMoreTokens()) {
                word.set(tokenizer.nextToken());
                context.write(word, one);
            }
        }
    }
}
```

```

    public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {
        public void reduce(Text key, Iterable<IntWritable> values, Context context) throws
IOException,
InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
            context.write(key, new IntWritable(sum));
        }
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Word Count");
    // Critical for Windows
    job.setJar("wordcount.jar");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(Map.class);
    job.setReducerClass(Reduce.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

PROCEDURE:

STEP 1: START HADOOP (Always first)

Open CMD as Administrator
cd c:\hadoop-3.3.0\sbin
hdfs namenode -format
start-all.cmd

STEP 2: Create the Java program

Open Notepad
Copy-paste your WordCount.java code
Save it as WordCount.java inside this folder C:\HadoopWork\src

STEP 3: Create input file (words.txt)

Go to: C:\HadoopWork\src
Create a new text file named:words.txt
Write this inside the file and save:
Hadoop, is cool.
Hadoop is great!
HADOOP is fast.

STEP 4: Create input directory in HDFS

```
cd c:\HadoopWork\src
Run:
hdfs dfs -mkdir /wc_input
This creates an input folder in HDFS.
```

STEP 5: Copy input file to HDFS

```
hdfs dfs -put words.txt /wc_input
```

STEP 6: Compile the Java program

```
javac -classpath "C:\hadoop-3.3.0\share\hadoop\common\*;C:\hadoop
3.3.0\share\hadoop\mapreduce\*;C:\hadoop-3.3.0\share\hadoop\common\lib\*"
WordCount.java
```

STEP 7: Create JAR file

```
jar cvf wordcount.jar WordCount*.class
```

STEP 8: Run WordCount job

```
hadoop jar wordcount.jar WordCount /wc_input /wc_output
```

```
Reduce input groups=7
Reduce shuffle bytes=110
Reduce input records=9
Reduce output records=7
Spilled Records=18
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=92
CPU time spent (ms)=1356
Physical memory (bytes) snapshot=569364480
Virtual memory (bytes) snapshot=822104064
Total committed heap usage (bytes)=460324864
Peak Map Physical memory (bytes)=325042176
Peak Map Virtual memory (bytes)=439410688
Peak Reduce Physical memory (bytes)=244322304
Peak Reduce Virtual memory (bytes)=382693376
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=54
File Output Format Counters
  Bytes Written=58

C:\HadoopWork\src>
```

STEP 9: View output

hdfs dfs -cat /wc_output/part-r-00000

```
C:\HadoopWork\src>hdfs dfs -cat /wc_output/part-r-00000
HADOOP 1
Hadoop 1
Hadoop, 1
cool. 1
fast. 1
great! 1
is 3
C:\HadoopWork\src>
```

RESULT:

Thus running a basic Word Count Map Reduce program to understand Map Reduce Paradigm was executed successfully.

**INSTALLATION OF HIVE ALONG WITH PRACTICE
EXAMPLES****AIM:**

To install HIVE along with practice examples.

PREREQUISITES:

1. Java Development Kit (JDK8) installed and the JAVA_HOME environment variable set.
2. Hadoop (version 2.9.2) installed and configured on your Windows system.
3. Apache Derby installed and DERBY_HOME environment variable set.

PROCEDURE:**Step 1: Download and Install Apache Derby**

- Download Apache Derby binary from:

<https://apache.root.lu/db/derby/db-derby-10.14.2.0/db-derby-10.14.2.0-bin.zip>

- Extract the ZIP file to C drive.
- Rename the extracted folder as derby.

Step 2: Download and Install Apache Hive

- Download Apache Hive binary from:

<https://archive.apache.org/dist/hive/hive-3.1.2/apache-hive-3.1.2-bin.tar.gz>

- Extract the file to C drive.
- Rename the extracted folder as hive.

Step 3: Environment Variable Configuration

- Set the following environment variables for proper execution of Hadoop, Hive, and Derby:

o JAVA_HOME → C:\Java\jdk1.8.0

o HADOOP_HOME → C:\hadoop-3.3.0

o DERBY_HOME → C:\derby

o HIVE_HOME → C:\hive

- Update the PATH variable by adding:

o %HADOOP_HOME%\bin

o %HADOOP_HOME%\sbin

- o %HIVE_HOME%\bin
- o %DERBY_HOME%\bin

step 4: Download Hive CMD Files

- Download Hive-cmd master ZIP from GitHub.
- Open the GitHub repository and click Code → Download ZIP.
- Extract the ZIP file.
- Copy all extracted .cmd files and paste them into:C:\hive

Step 5: Copy Derby Library Files

- Copy all files from:C:\derby\lib
- Paste them into:C:\hive\lib

step 6:

- Navigate to:C:\hadoop-3.3.0\share\hadoop\common\lib
- Copy the file:guava-27.0-jre.jar
- Paste it into:C:\hive\lib
- Delete the older file:guava-19.0.jar

Step 7:

Create hive-site.xml inside hive/conf and paste the following code in it:

```
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>

  <property>
    <name>javax.jdo.option.ConnectionURL</name>
    <value>jdbc:derby:C:/hive/metastore_db;create=true</value>
    <description>Derby database location on your local C drive</description>
  </property>

  <property>
    <name>javax.jdo.option.ConnectionDriverName</name>
```

```
<value>org.apache.derby.jdbc.EmbeddedDriver</value>
</property>

<property>
  <name>javax.jdo.option.ConnectionUserName</name>
  <value>APP</value>
</property>

<property>
  <name>javax.jdo.option.ConnectionPassword</name>
  <value>mine</value>
</property>

<property>
  <name>datanucleus.identifier.schema</name>
  <value>APP</value>
</property>

<property>
  <name>hive.metastore.schema.validation</name>
  <value>false</value>
</property>

<property>
  <name>datanucleus.schema.autoCreateAll</name>
  <value>true</value>
</property>

<property>
  <name>hive.metastore.warehouse.dir</name>
  <value>/user/hive/warehouse</value>
</property>
```

```
<property>
  <name>hive.exec.scratchdir</name>
  <value>/tmp/hive</value>
</property>

<property>
  <name>hive.exec.local.scratchdir</name>
  <value>C:/hive/scratchdir</value>
</property>

<property>
  <name>hive.querylog.location</name>
  <value>C:/hive/scratchdir</value>
</property>
<property>
  <name>hive.scratch.dir.permission</name>
  <value>733</value>
</property>
</configuration>
```

Step 8:

- Open Command Prompt.
- Navigate to:C:\hive\bin
- Execute the command:

```
hive --service schematool -dbType derby -initSchema --verbose
```



```

No rows affected (0.026 seconds)
0: jdbc:derby:C:/hive/metastore_db>
0: jdbc:derby:C:/hive/metastore_db> CREATE UNIQUE INDEX "APP"."UNIQUE_SCHEMA_VERSION" ON "APP"."SCHEMA_VERSION" ("SCHEMA_ID", "VERSION")
No rows affected (0.006 seconds)
0: jdbc:derby:C:/hive/metastore_db>
0: jdbc:derby:C:/hive/metastore_db> CREATE TABLE REPL_TXN_MAP ( RTM_REPL_POLICY varchar(256) NOT NULL, RTM_SRC_TXN_ID bigint NOT NULL, RTM_TARGET_TXN_ID bigint NOT NULL, PRIMARY KEY (RTM_REPL_POLICY, RTM_SRC_TXN_ID) )
No rows affected (0.012 seconds)
0: jdbc:derby:C:/hive/metastore_db>
0: jdbc:derby:C:/hive/metastore_db> CREATE TABLE "APP"."RUNTIME_STATS" ( "RS_ID" bigint primary key, "CREATE_TIME" integer not null, "WEIGHT" integer not null, "PAYLOAD" BLOB )
No rows affected (0.013 seconds)
0: jdbc:derby:C:/hive/metastore_db>
0: jdbc:derby:C:/hive/metastore_db> CREATE INDEX IDX_RUNTIME_STATS_CREATE_TIME ON RUNTIME_STATS(CREATE_TIME)
No rows affected (0.006 seconds)
0: jdbc:derby:C:/hive/metastore_db>
0: jdbc:derby:C:/hive/metastore_db> INSERT INTO "APP"."VERSION" (VER_ID, SCHEMA_VERSION, VERSION_COMMENT) VALUES (1, '3.1.0', 'Hive release version 3.1.0')
1 row affected (0.01 seconds)
0: jdbc:derby:C:/hive/metastore_db>
0: jdbc:derby:C:/hive/metastore_db> !closeall
Closing: 0: jdbc:derby:C:/hive/metastore_db;create=true
beeline>
beeline>
beeline> Initialization script completed
schemaTool completed

C:\Windows\System32>

```

Step 9 Start Hadoop Services

- Open Command Prompt.
- Navigate to:C:\hadoop-3.3.0\sbin
- Start Hadoop services using:

```
start-dfs.cmd
start-yarn.cmd
```

Step 10: Start Derby Network Server

- Open a new Command Prompt.
- Navigate to:C:\derby\bin
- Start the Derby server using the cmd:startNetworkServer -h 0.0.0.0

step 11:

- Open a new Command Prompt.
- Navigate to:C:\hive\bin
- Start Hive using the cmd:hive

- Execute the following Hive commands one by one:
 - set hive.exec.mode.local.auto=true;
 - CREATE DATABASE ipdir;
 - SHOW DATABASES;
 - USE ipdir;
 - CREATE TABLE student(id int, name string, age int) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';
 - INSERT INTO student VALUES (1, 'John Doe', 25);
 - SELECT * FROM student;

```

re.q1.metadata.sessionhiveclient ugi=SriJainthu (auth:SIMPLE) retries=1 delay=1 lifetime=0
2026-03-17 14:50:33,207 INFO metastore.HiveMetaStore: 0: create_database: Database(name:ipdir, description:null, loc
hUri:hdfs://localhost:9000/user/hive/warehouse/ipdir.db, parameters:null, ownerName:SriJainthu, ownerType:USER, cata
ame:hive)
2026-03-17 14:50:33,207 INFO HiveMetaStore.audit: ugi=SriJainthu ip=unknown-ip-addr cmd=create_database:
abase(name:ipdir, description:null, locationUri:hdfs://localhost:9000/user/hive/warehouse/ipdir.db, parameters:null,
erName:SriJainthu, ownerType:USER, catalogName:hive)
2026-03-17 14:50:33,219 WARN metastore.ObjectStore: Failed to get database hive.ipdir, returning NoSuchObjectExcepti
2026-03-17 14:50:33,254 INFO utils.FileUtils: Creating directory if it doesn't exist: hdfs://localhost:9000/user/hiv
arehouse/ipdir.db
2026-03-17 14:50:33,301 INFO ql.Driver: Completed executing command(queryId=SriJainthu_20260317145032_3f4cdeda-a090-
-be26-ebc0ab9396d8); Time taken: 0.215 seconds
OK
2026-03-17 14:50:33,301 INFO ql.Driver: OK
2026-03-17 14:50:33,301 INFO ql.Driver: Concurrency mode is disabled, not creating a lock manager
Time taken: 0.963 seconds
2026-03-17 14:50:33,301 INFO CliDriver: Time taken: 0.963 seconds
2026-03-17 14:50:33,301 INFO conf.HiveConf: Using the default value passed in for log id: 2beaad64-daf9-4f1d-92df-5b
dc756e
2026-03-17 14:50:33,301 INFO session.SessionState: Resetting thread name to main
hive>

```

RESULT:

Thus, the installation of HIVE along with practice examples was executed successfully

AIM:

To install hbase along with practice examples.

PROCEDURE:**Step 1: Download and Extract HBase**

- Download **hbase-2.2.5-bin.tar.gz** from the Apache archive
https://archive.apache.org/dist/hbase/2.2.5/?utm_source=chatgpt.com.
- Extract the file to the **C drive** and rename the folder as: C:\hbase-2.2.5
- Create the following empty directories for data storage and ZooKeeper:

C:\hbase-2.2.5\data

C:\hbase-2.2.5\zookeeper

Step-2: Modify HBase Startup Script

Open the HBase startup script file: C:\hbase-2.2.5\bin\hbase.cmd

Look for this specific line:

```
set java_arguments=%HEAP_SETTINGS% %HBASE_OPTS% -classpath
```

```
"%CLASSPATH%" %CLASS% %hbase-command-arguments%
```

Remove %HEAP_SETTINGS% so the line looks exactly like this:

```
set java_arguments=%HBASE_OPTS% -classpath "%CLASSPATH%" %CLASS% %hbase-  
command-arguments%
```

Step 3: Configure hbase-env.cmd

- Open the file: C:\hbase-2.2.5\conf\hbase-env.cmd
- Add the following lines at the end of the file:

```
set JAVA_HOME=C:\Java\jdk-1.8
```

```
set HBASE_MANAGES_ZK=true
```

```
set HBASE_CLASSPATH=%HBASE_HOME%\lib\client-facing-thirdparty\*
```

Step 4: Configure hbase-site.xml

- Open the file: C:\hbase-2.2.5\conf\hbase-site.xml
- Replace the <configuration> section with:

```
<configuration>
```

```
<property>
```

```
<name>hbase.rootdir</name>
```

```

    <value>file:///C:/hbase-2.2.5/data</value>
  </property>
  <property>
<name>hbase.zookeeper.property.dataDir</name>
  <value>C:/hbase-2.2.5/zookeeper</value>
  </property>
  <property>
    <name>hbase.cluster.distributed</name>
<value>false</value>
  </property>
  <property>
    <name>hbase.unsafe.stream.capability.enforce</name>
    <value>false</value>
  </property>
</configuration>

```

Step 5: Environment Variable Setup

- Set the following system environment variable:
 - **HBASE_HOME** → C:\hbase-2.2.5
- Add the following path to the **PATH** variable: %HBASE_HOME%\bin

Step 6: Start HBase Services

- Open **Command Prompt as Administrator**.
- Navigate to: cd C:\hbase-2.2.5\bin
- Start HBase using: start-hbase.cmd

Step 7: Use HBase Shell

- In the same Command Prompt, open HBase shell: hbase shell
- Create a table: create 'emp', 'personal_data'
- Insert data into the table:


```

put 'emp', '1', 'personal_data:name', 'raju'
put 'emp', '1', 'personal_data:city', 'hyderabad'

```
- Display the stored data: scan 'emp'

Step 8: Verify Using HBase Web UI

- Open a web browser (Chrome or Edge).
- Go to: <http://localhost:16010>
- The **HBase Master Status** page will be displayed.

- Under **User Tables**, the emp table will be visible, confirming successful setup.

The screenshot shows the Apache HBase master-status page at localhost:16010/master-status. The 'Tables' section is active, showing a list of tables. The 'emp' table is visible in the 'default' namespace, with a state of 'ENABLED' and 1 region.

Namespace	Name	State	Regions								Description
			OPEN	OPENING	CLOSED	CLOSING	OFFLINE	FAILED	SPLIT	Other	
default	emp	ENABLED	1	0	0	0	0	0	0	0	'emp', {NAME => 'personal_data'}

RESULT:

Thus the installation of hbase along with practice examples was executed successfully

AIM:

To install hbase along with practice examples.

PROCEDURE:**Step 1: Download and Setup Thrift**

Since we are on Windows, we use the pre-compiled binary instead of building from source.

1. **Download:** Get thrift-0.13.0.exe from the official Apache archive.
2. **Move & Rename:** Move the file to C:\hbase-2.2.5\bin and rename it to **thrift.exe**.
3. **Environment Variable:** Add C:\hbase-2.2.5\bin to your System **Path** variables so the command works globally.

Step 2: Verify Installation

Open a new Command Prompt and verify the version:

DOS

thrift -version

Expected Output: Thrift version 0.13.0

Step 3: Create a Thrift Definition File

Create a practice definition file to tell Thrift what kind of code to generate.

1. Open Notepad and paste:

```
Thrift
service HBaseTest {
    void ping(),
    string getStatus()
}
```

2. Save it in C:\hbase-2.2.5\bin as **lab.thrift**.

Step 4: Practice Example (Code Generation)

Generate the Python "stubs" (the client-side bridge code).

1. Run the compiler command:

```
thrift --gen py lab.thrift
```

2. Verify the generated files:

```
dir gen-py\lab
```

Expected Result: You will see HBaseTest.py, constants.py, and ttypes.py. These files allow Python to talk to the Thrift server.

Step 5: Start the HBase Thrift Server

Now, open the "bridge" that connects external code to your database.

1. Run the Thrift service:

DOS

hbase thrift start

Expected Result: The terminal will show starting HBase ThreadPool ThriftServer on port 9090.

```
p.lib.StaticUserWebFilter$StaticUserFilter) to context logs
2026-03-17 14:58:24,350 INFO [main] http.HttpServer: ASYNC_PROFILER_HOME environment variable and async.profiler.home
system property not specified. Disabling /prof endpoint.
2026-03-17 14:58:24,373 INFO [main] http.HttpServer: Jetty bound to port 9095
2026-03-17 14:58:24,373 INFO [main] server.Server: jetty-9.3.27.v20190418, build timestamp: 2019-04-18T23:41:38+05:30,
git hash: d3e249f86955d04bc646bb620905b7c1bc596a8d
2026-03-17 14:58:24,404 INFO [main] http.SecurityHeadersFilter: Added security headers filter
2026-03-17 14:58:24,425 INFO [main] handler.ContextHandler: Started o.e.j.s.ServletContextHandler@7651218e{/logs,file:
//C:/hbase-2.2.5/logs,AVAILABLE}
2026-03-17 14:58:24,425 INFO [main] http.SecurityHeadersFilter: Added security headers filter
2026-03-17 14:58:24,425 INFO [main] handler.ContextHandler: Started o.e.j.s.ServletContextHandler@24faea88{/static,fil
:///C:/hbase-2.2.5/hbase-webapps/static/,AVAILABLE}
2026-03-17 14:58:24,707 INFO [main] webapp.StandardDescriptorProcessor: NO JSP Support for /, did not find org.eclipse
jetty.jsp.JettyJspServlet
2026-03-17 14:58:24,723 INFO [main] http.SecurityHeadersFilter: Added security headers filter
2026-03-17 14:58:24,743 INFO [main] handler.ContextHandler: Started o.e.j.w.WebAppContext@285f09de[/,file:///C:/hbase-
.2.5/hbase-webapps/thrift/,AVAILABLE]{file:/C:/hbase-2.2.5/hbase-webapps/thrift}
2026-03-17 14:58:24,759 INFO [main] server.AbstractConnector: Started ServerConnector@1a6f5124[HTTP/1.1,[http/1.1]]{0.
.0.0:9095}
2026-03-17 14:58:24,759 INFO [main] server.Server: Started @2727ms
2026-03-17 14:58:24,763 INFO [main] thrift.ThriftServer: starting HBase ThreadPool Thrift server on /0.0.0.0:9090
```

RESULT:

Thus the installation of hbase along with practice examples was executed successfully.

EX NO:7

DATE:

PRACTICE IMPORTING AND EXPORTING DATA FROM VARIOUS DATABASES

AIM:

Practice importing and exporting data from various databases.

PREREQUISITES:

1. Java JDK 8
2. Hadoop (3.3.x)
3. Hive (3.1.x)
4. MySQL Server 5.7.44 (Community Installer)
5. MySQL Connector/J (JDBC)
6. Sqoop 1.4.6

PROCEDURE:

Step 1:Ensure java 8 and start Hadoop:

Step 2:Install MySQL Server-mysql-installer.5.x.x.

PASSWORD root123

after installation, MySQL is usually in:

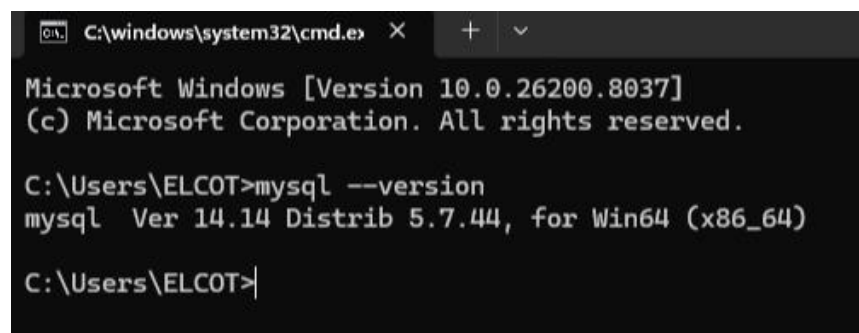
C:\Program Files\MySQL\MySQL Server 5.0\bin

Add to PATH:

C:\Program Files\MySQL\MySQL Server 5.0\bin

mysql --version

Expected output:



```
C:\windows\system32\cmd.exe X + v
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ELCOT>mysql --version
mysql Ver 14.14 Distrib 5.7.44, for Win64 (x86_64)

C:\Users\ELCOT>
```



```
mysql -u root -p
```

Enter password:

```
root123
```

You should enter MySQL prompt:

```
mysql>
```

Step 3:Install sqoop and extract in C

Add the environment variables;

Variable Name : SQOOP_HOME

Variable Value: C:\sqoop

Path:

```
%SQOOP_HOME%\bin
```

```
sqoop version
```

Expected Output

```
Sqoop 1.4.6
```

STEP 4: INSTALL MySQL CONNECTOR

Copy ONLY this .jar file

```
mysql-connector-java-5.1.42.jar
```

Paste it into:

```
C:\sqoop\lib
```

Step 5:Check MySQL is running

Open new Command Prompt

Run:

```
mysql --version
```

Expected:

```
mysql Ver 5.0.x for Win64
```

Then:

```
mysql -u root -p
```

If MySQL prompt opens → SUCCESS

```
CREATE USER 'sqoop'@'localhost' IDENTIFIED BY 'sqooppassword';
```

```
GRANT ALL PRIVILEGES ON *.* TO 'sqoop'@'localhost';
```

```
FLUSH PRIVILEGES;
```

```
USE employees;
```

```
SHOW TABLES;
```

```
CREATE DATABASE employees;
```

```
USE employees;
```

```
CREATE TABLE practice_table (
```

```
id INT PRIMARY KEY,
```

```
name VARCHAR(20)
```

```
);
```

```
INSERT INTO practice_table VALUES (1, 'IT'), (2, 'Finance');
```

```
mysql> CREATE DATABASE employees;
Query OK, 1 row affected (0.00 sec)

mysql> USE employees;
Database changed
mysql> CREATE TABLE practice_table (
  -> id INT PRIMARY KEY,
  -> name VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO practice_table VALUES (1, 'IT'), (2, 'Finance');
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

Load the Data: Use the dataset from the GitHub link. If you're using the "Employees" sample database, you can usually load it by running:

```
mysql -u root -p < employees.sql
```

Phase 3: The Data Transfer (The Core of the Lab)

1. Verify Connection

Before importing, check if Sqoop can see your MySQL databases:

```
sqoop list-databases --connect "jdbc:mysql://localhost/?useSSL=false" --username sqoop --password sqooppassword
```

1. Re-Import Data to HDFS

First, we have to move the data from MySQL back into HDFS so there is actually something there to "cat."

DOS

```
sqoop import ^  
  
--connect "jdbc:mysql://localhost/employees?useSSL=false" ^  
  
--username sqoop ^  
  
--password sqooppassword ^  
  
--table practice_table ^  
  
--target-dir /user/user/practice_data ^  
  
--delete-target-dir ^  
  
--m 1
```

```

C:\Windows\System32>sqoop import ^
More? --connect "jdbc:mysql://localhost/employees?useSSL=false" ^
More? --username sqoop ^
More? --password sqooppassword ^
More? --table practice_table ^
More? --target-dir /user/user/practice_data ^
More? --delete-target-dir ^
More? --m 1
Warning: HCAT_HOME not set
Warning: HCATALOG_HOME does not exist HCatalog imports will fail.
Please set HCATALOG_HOME to the root of your HCatalog installation.
Warning: ACCUMULO_HOME not set.
Warning: ZOOKEEPER_HOME not set.
Warning: ACCUMULO_HOME does not exist Accumulo imports will fail.
Please set ACCUMULO_HOME to the root of your Accumulo installation.
Warning: ZOOKEEPER_HOME does not exist Accumulo imports will fail.
Please set ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-03-18 21:19:09,013 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6
2026-03-18 21:19:09,047 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-03-18 21:19:09,228 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
2026-03-18 21:19:09,228 INFO tool.CodeGenTool: Beginning code generation
2026-03-18 21:19:09,605 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `practice_table` AS t LIMIT 1
2026-03-18 21:19:09,637 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `practice_table` AS t LIMIT 1
2026-03-18 21:19:09,653 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is C:\hadoop
Note: \tmp\sqoop-ELCOT\compile\de45e4cbb863975771b6f0cc7bbc1cd0\practice_table.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-03-18 21:19:12,924 INFO orm.CompilationManager: Writing jar file: \tmp\sqoop-ELCOT\compile\de45e4cbb863975771b6f0cc7bbc1cd0\practice_table.jar
2026-03-18 21:19:14,067 INFO tool.ImportTool: Destination directory /user/user/practice_data is not present, hence not deleting.
2026-03-18 21:19:14,067 WARN manager.MySQLManager: It looks like you are importing from mysql.
2026-03-18 21:19:14,067 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
2026-03-18 21:19:14,067 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
2026-03-18 21:19:14,067 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
2026-03-18 21:19:14,067 INFO mapreduce.ImportJobBase: Beginning import of practice_table

```

2. Verify the HDFS File (The "Cat" Step)

Now that the import has finished, try your command again. It should now show you the CSV-formatted data.

DOS

`hadoop fs -cat /user/user/practice_data/part-m-00000`

```

C:\Windows\System32>hadoop fs -cat /user/user/practice_data/part-m-00000
1,IT
2,Finance
C:\Windows\System32>_

```

3. Verify in Hive

To check this in Hive, you first need to tell Hive where the data is. Open the Hive shell and run:

SQL

-- Create the table structure in Hive

`CREATE TABLE IF NOT EXISTS hive_practice_table (`

`id INT,`

`name STRING,`

`designation STRING`

)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ',';

-- Load the data from your Sqoop HDFS directory

-- Note: 'LOAD DATA' moves the file, so the HDFS folder will be empty after this

LOAD DATA INPATH '/user/user/practice_data/part-m-00000' INTO TABLE
hive_practice_table;

-- Verify the data

SELECT * FROM hive_practice_table LIMIT 5;

3. Export from HDFS back to MySQL

Now that HDFS is populated again, you can finally complete the export to your "export" table in MySQL. Run this in your Command Prompt:

First, in MySQL:

SQL

USE employees;

CREATE TABLE practice_table_export AS SELECT * FROM practice_table WHERE 1=0;

DOS

sqoop export ^

--connect "jdbc:mysql://localhost/employees?useSSL=false" ^

--username sqoop ^

--password sqooppassword ^

--table practice_table_export ^

--export-dir /user/hive/warehouse/hive_practice_table ^

--input-fields-terminated-by "," ^

--m 1

```

C:\Windows\System32>sqoop export ^
More? --connect "jdbc:mysql://localhost/employees?useSSL=false" ^
More? --username sqoop ^
More? --password sqooppassword ^
More? --table practice_table_export ^
More? --export-dir /user/hive/warehouse/hive_practice_table ^
More? --input-fields-terminated-by "," ^
More? --m 1
Warning: HCAT_HOME not set
Warning: HCATALOG_HOME does not exist HCatalog imports will fail.
Please set HCATALOG_HOME to the root of your HCatalog installation.
Warning: ACCUMULO_HOME not set.
Warning: ZOOKEEPER_HOME not set.
Warning: ACCUMULO_HOME does not exist Accumulo imports will fail.
Please set ACCUMULO_HOME to the root of your Accumulo installation.
Warning: ZOOKEEPER_HOME does not exist Accumulo imports will fail.
Please set ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-03-19 16:48:45,215 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6
2026-03-19 16:48:45,256 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-03-19 16:48:45,415 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
2026-03-19 16:48:45,415 INFO tool.CodeGenTool: Beginning code generation
2026-03-19 16:48:45,801 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `practice_table_export` AS t LIMIT 1
2026-03-19 16:48:45,848 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `practice_table_export` AS t LIMIT 1
2026-03-19 16:48:45,860 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is C:\hadoop
Note: \tmp\sqoop-ELCOT\compile\8ec24d5296e23c1259d095a2a721eb2b\practice_table_export.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-03-19 16:48:49,366 INFO orm.CompilationManager: Writing jar file: \tmp\sqoop-ELCOT\compile\8ec24d5296e23c1259d095a2a721eb2b\practice
2026-03-19 16:48:49,399 INFO mapreduce.ExportJobBase: Beginning export of practice_table_export
2026-03-19 16:48:49,399 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-03-19 16:48:49,544 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2026-03-19 16:48:50,741 INFO Configuration.deprecation: mapred.reduce.tasks.speculative.execution is deprecated. Instead, use mapreduce.
2026-03-19 16:48:50,757 INFO Configuration.deprecation: mapred.map.tasks.speculative.execution is deprecated. Instead, use mapreduce.map
2026-03-19 16:48:50,757 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2026-03-19 16:48:50,870 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-19 16:48:56,770 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ELCOT/.staging/j
2026-03-19 16:48:58,808 INFO input.FileInputFormat: Total input files to process : 2
2026-03-19 16:48:58,816 INFO input.FileInputFormat: Total input files to process : 2
2026-03-19 16:48:58,903 INFO mapreduce.JobSubmitter: number of splits:1

```

4. Final Verification

To prove the cycle is complete, go to your MySQL command line and check the results:

SQL

USE employees;

Select * from practice_table;

```

mysql> USE employees;
Database changed
mysql> Select * from practice_table;
+----+-----+
| id | name  |
+----+-----+
| 1  | IT    |
| 2  | Finance |
+----+-----+
2 rows in set (0.00 sec)

mysql> |

```

RESULT:

Thus practice importing and exporting data from various databases was executed successfully.